Neel Gali

Software Engineer
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Summary

- Designed and owned low-latency backend services for ad analytics workflows, enabling near real-time aggregation of impression, click, and conversion metrics at scale.
- Built high-throughput data processing pipelines in Java and Python to ingest and aggregate millions of events per day, optimizing hot paths to reduce end-to-end latency by 30
- Implemented custom in-memory data structures to support fast rolling-window aggregations required for live Ads reporting and monitoring use cases.
- Led performance profiling and debugging efforts using structured logging, metrics, and trace analysis to identify bottlenecks in CPU, memory, and I/O-heavy services.
- Improved system reliability by introducing health checks, circuit breakers, and automated regression tests, significantly reducing production incidents.
- Partnered with product and analytics teams to translate Ads performance requirements into scalable backend designs, clearly communicating trade-offs around accuracy, freshness, and cost.

Skills

Languages: Python, Java, C++
Core CS: Data Structures & Algorithms, OOP, Multithreading, Memory Management
Systems: Distributed Systems, Large-Scale Data Processing, Performance Optimization
Debugging & Reliability: Logging, Monitoring, Root Cause Analysis, Incident Response
Testing: Unit Testing, Integration Testing, Load & Regression Testing
Data & Visualization: SQL, internal analytics tooling, metrics dashboards
Databases & Data Management: SQL (MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB)
Accessibility: Inclusive UI patterns, performance-aware backend support
Tools: Git, Bazel, Linux, CI/CD pipelines

Professional Experience

Epic Solutions *Jun 2025 – Present*
Software Engineer II
Project: Consumer Technology

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Samay Systems LLC *July 2024 – Jun 2025*
Software Engineer I
Project: SaaS Platform

- Developed data-intensive backend services in Python and C++ powering analytics dashboards and customer-facing reporting features.
- Applied strong data structures and algorithms fundamentals to optimize aggregation, filtering, and ranking logic for large datasets.
- Implemented incremental computation strategies to avoid full recomputation of metrics, improving system responsiveness under load.
- Assisted in diagnosing and resolving complex production issues in distributed services by analyzing logs, metrics, and execution traces.
- Strengthened code quality and long-term maintainability by writing comprehensive unit and integration tests and participating in structured code reviews.
- Contributed to system design discussions, focusing on scalability limits, failure modes, and graceful degradation.

Monarch Tractor *Jun 2023 – Aug 2023*
SDE Intern
Project: HR workflow Automation

- Built backend automation services to streamline employee and device lifecycle workflows, replacing manual processes with reliable programmatic systems.
- Integrated multiple third-party APIs into a unified service layer, ensuring consistent data flow and robust error handling.
- Designed idempotent background jobs and monitoring hooks to guarantee correctness and observability in automated workflows.
- Gained hands-on experience operating production systems, debugging failures, and improving reliability through logging and metrics.

ZeMoSo Technologies *Jul 2021 – Aug 2022*
Associate Developer
Project: Biotech LabOps Automation

- Designed and implemented backend APIs supporting high-volume application workflows, emphasizing correctness, performance, and security.
- Built data-backed features that required efficient querying, transformation, and response-time guarantees under concurrent access.
- Worked closely with frontend engineers to ensure backend systems supported accessible, responsive user experiences, including predictable latency and clear error semantics.
- Participated in on-call support and post-incident analysis, strengthening skills in debugging, root-cause analysis, and long-term fixes.
- Participated in system debugging and production support, strengthening operational reliability.

Projects

High-Performance Data Aggregation Engine

- Designed a multi-threaded data aggregation service in C++, reducing processing time by 40%.
- Implemented custom data structures to support efficient lookups and updates under heavy load.

Agent-based Analytics Platform – Samay Systems

- Built backend services to collect, validate, and analyze large volumes of structured and semi-structured data for analytics and reporting.

- Designed query and retrieval logic optimized for low-latency access patterns required by interactive dashboards.
- Focused on system correctness, auditability, and explainability of outputs.

System Health & Monitoring Framework– Zemoso

- Built a Python-based monitoring layer to track latency, error rates, and throughput.
- Enabled faster incident diagnosis and improved overall system stability.

Education

Master's in Computer Science, Oklahoma State University, Stillwater, OK *May 2024*
Bachelor's in Computer Science, JNTU Hyderabad, India *Jun 2021*